



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

08-04-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,D	B,C	B,C	A,B	A,B,C,D	A,B
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	D	A	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.20	1.00	0.22 to 0.24	3.10	1.50 to 1.70	10.40 to 10.60
SECTION-II (ii)	Q.	7	8	9			
	A.	3	3	6			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,B,C,D	A,B,D	A,C,D	A,B,C,D	A,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	D	A	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	330.00 to 342.00	0.75 to 0.80	4.00	2.00	190.60 to 190.90	143.50 to 143.60
SECTION-II (ii)	Q.	7	8	9			
	A.	5	1	6			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,D	B,C	A,B,D	B,C	B,C	A,B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	C	A	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	55.00	66.00	0.12 to 0.13	0.04	0.00	6.28 to 6.30
SECTION-II (ii)	Q.	7	8	9			
	A.	5	4	8			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,D)

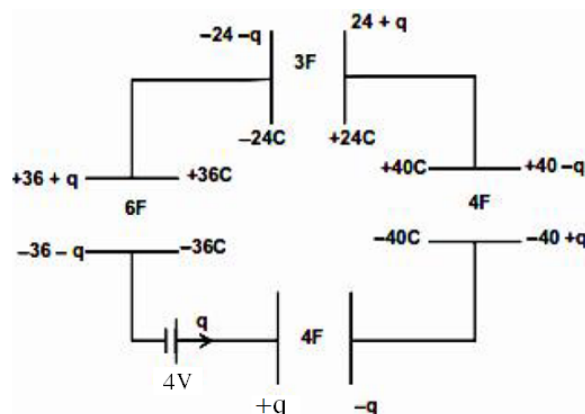
By applying Kirchhoff's Law

$$-4 + \frac{q}{4} + \frac{-40 + q}{4} + \frac{24 + q}{3} + \frac{36 + q}{6} = 0$$

$$-4 + \frac{q}{4} - 10 + \frac{q}{4} + 8 + \frac{q}{3} + 6 + \frac{q}{6} = 0$$

$$\frac{q}{4} + \frac{q}{4} + \frac{q}{3} + \frac{q}{6} = 0 \Rightarrow q = 0$$

Hence, there is no flow of charge



2. Ans (B,C)

The electrostatic force between the hemispherical shells is

$$F = \frac{\sigma_1 \sigma_2 \pi R^2}{2\epsilon_0} = \frac{Q}{2\pi R^2} \frac{2Q}{8\pi R^2} \frac{\pi R^2}{2\epsilon_0}$$

$$= \frac{Q^2}{16\pi\epsilon_0 R^2}$$

The net electric potential at the centre 'O' of the shell is

$$V_o = \frac{KQ}{R} + \frac{2kQ}{2R} = \frac{2KQ}{R} = \frac{Q}{2\pi\epsilon_0 R}$$

3. Ans (B,C)

$$V_{CM} = \omega \sqrt{R^2 - \frac{3R^2}{4}} = \omega \frac{R}{2} \text{ (from pure rolling)}$$

$$mgh = \frac{1}{2}mv_{cm}^2 + \frac{1}{2} \frac{2}{5}mR^2\omega^2 \text{ (from}$$

conservation of energy)

$$\therefore V_{CM} = \sqrt{\frac{10gh}{13}}$$

4. Ans (A,B)

$$\delta_{net} = \delta_1 + \delta_2 + \dots + \delta_n$$

Where all have same magnitude and alternate deviations are in opposite directions.

5. Ans (A,B,C,D)

I	II
$P_1 : V_1$	$P_1 : V_2$
T_1	T_1

The ΔU of the system should be zero.

$$\text{or } n_1 C_V (\Delta T_1) + n_2 C_V (\Delta T_2) = 2$$

$$\therefore T_1 = T$$

equating no. of moles,

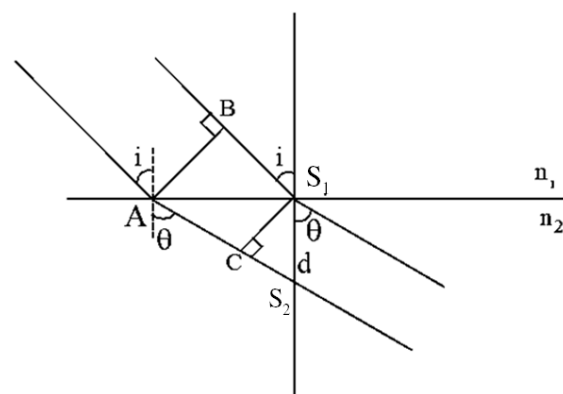
$$\frac{PV}{RT} = \frac{P_1 V_1}{RT} \text{ and } \frac{(2P)(2V)}{RT} = \frac{P_1 V_2}{RT}$$

$$PV = P_1 V_1 \text{ and } 4PV = P_1 V_2$$

$$\therefore 4V_1 = V_2$$

$$V_1 = \frac{3V}{5} : V_2 = \frac{12V}{5} \text{ and } P_1 = \frac{5P}{3}$$

6. Ans (A,B)



$$\Delta x = n_2 AC - n_1 BS_1$$

$$AC = d \sin \theta \tan \theta$$

$$BS_1 = \frac{d \sin \theta \sin i}{\cos \theta}$$

$$\text{By Snell's Law } n_1 \sin i = n_2 \sin \theta \Rightarrow \Delta x = 0$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (B)

Total charge in an element = $Rd\theta \times \lambda_0 \sin \theta$

On revolving this ring we get a sphere of charge

$$\text{density} = \frac{Rd\theta\lambda_0 \sin \theta}{2\pi R \sin \theta Rd\theta} \text{ which is a constant.}$$

Basic concept of potential $V_A - V_B = E.R$

8. Ans (D)

Total charge in an element = $Rd\theta \times \lambda_0 \sin \theta$

On revolving this ring we get a sphere of charge

$$\text{density} = \frac{Rd\theta\lambda_0 \sin \theta}{2\pi R \sin \theta Rd\theta} \text{ which is a constant.}$$

Basic concept of potential $V_A - V_B = E.R$

9. Ans (A)

$$\phi = Li = \frac{\mu_0 m}{2\pi r^3} \times \pi a^2$$

$$\Rightarrow i = \frac{\mu_0 m \pi a^2}{2\pi r^3 L} \Rightarrow i \propto \frac{m}{r^3}$$

$$m' = \pi a^2 i = \frac{\mu_0 m \pi^2 a^4}{2\pi r^3 L}$$

$$F = \frac{km^2 \pi^2 a^4}{2\pi r^7 L}$$

$$W = \int F dr \propto \int \frac{m^2 dr}{r^7}$$

$$W \propto \frac{m^2}{r^6}$$

10. Ans (C)

$$\phi = Li = \frac{\mu_0 m}{2\pi r^3} \times \pi a^2$$

$$\Rightarrow i = \frac{\mu_0 m \pi a^2}{2\pi r^3 L} \Rightarrow i \propto \frac{m}{r^3}$$

$$m' = \pi a^2 i = \frac{\mu_0 m \pi^2 a^4}{2\pi r^3 L}$$

$$F = \frac{k m^2 \pi^2 a^4}{2\pi r^7 L}$$

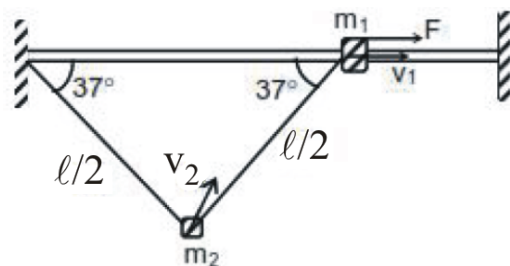
$$W = \int F dr \propto \int \frac{m^2 dr}{r^7}$$

$$W \propto \frac{m^2}{r^6}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (1.20)



$$v_1 \cos \theta = v_2 \sin 2\theta$$

$$v_1 = 2v_2 \sin \theta$$

$$v_1 = 2v_2 \left(\frac{3}{5} \right) \Rightarrow v_2 = \frac{5}{6} v_1$$

Using work energy theorem

$$F \ell (\cos \theta - \cos \theta_0) - m_2 g \frac{\ell}{2} (\sin \theta_0 - \sin \theta)$$

$$= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$20\ell \left(\frac{4}{5} - \frac{3}{5} \right) - 20 \times \frac{\ell}{2} \left(\frac{4}{5} - \frac{3}{5} \right)$$

$$= \frac{1}{2} \times 5v_1^2 + \frac{1}{2} \times 2 \times \frac{25}{36} v_1^2$$

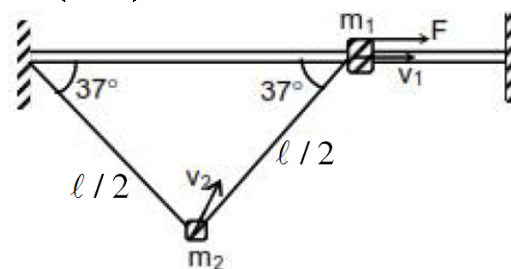
$$2(4\ell - 2\ell) = \left(5 + \frac{50}{36} \right) v_1^2$$

$$\Rightarrow v_1 = \sqrt{\frac{4 \times 2.3 \times 36}{230}} = 1.2 \text{ m/s}$$

$$v_1 = 1.2 \text{ m/s}$$

$$v_2 = \frac{5}{6} v_1 = 1 \text{ m/s}$$

2. Ans (1.00)



$$v_1 \cos \theta = v_2 \sin 2\theta$$

$$v_1 = 2v_2 \sin \theta$$

$$v_1 = 2v_2 \left(\frac{3}{5} \right) \Rightarrow v_2 = \frac{5}{6} v_1$$

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$$20\ell \left(\frac{4}{5} - \frac{3}{5} \right) - 20 \times \frac{\ell}{2} \left(\frac{4}{5} - \frac{3}{5} \right)$$

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$$v_1 = 1.2 \text{ m/s}$$

$$v_2 = \frac{5}{6} v_1 = 1 \text{ m/s}$$

3. Ans (0.22 to 0.24)

Relative velocity of the block after two collision

$$= \frac{v}{4} \text{ towards right.}$$

From conservation of momentum

$$mv = m \left(\frac{v}{4} + v_w \right) + 32mv_w$$

$$\text{So, } v_w = \frac{v}{44} = \frac{10}{44} = 0.23 \text{ m/s}$$

Velocity of the wagon, when the block is at rest

$$\text{respect to ground} = \frac{v}{32} \text{ m/s}$$

So, time taken

$$= \frac{d}{v} + \frac{d}{\frac{v}{2}} + \frac{d}{\frac{v}{4}} + \frac{d}{\frac{v}{8}} + \frac{d}{\frac{v}{16}} = \frac{31d}{v} = 3.10 \text{ sec}$$

4. **Ans (3.10)**

Relative velocity of the block after two

collision = $\frac{v}{4}$ towards right.

From conservation of momentum

$$mv = m \left(\frac{v}{4} + v_w \right) + 32mv_w$$

$$\text{So, } v_w = \frac{v}{44} = \frac{10}{44} = 0.23 \text{ m/s}$$

Velocity of the wagon, when the block is at rest

$$\text{respect to ground} = \frac{v}{32} \text{ m/s}$$

So, time taken

$$= \frac{d}{v} + \frac{d}{\frac{v}{2}} + \frac{d}{\frac{v}{4}} + \frac{d}{\frac{v}{8}} + \frac{d}{\frac{v}{16}} = \frac{31d}{v} = 3.10 \text{ sec}$$

5. **Ans (1.50 to 1.70)**

Using conservation of energy

$$(m \times 540) + m \times 1 \times (100 - 20)$$

$$= (m \times 80) + M \times 1 \times (20 - 0)$$

$$\Rightarrow m = 6.2 \text{ m}$$

$$(0.2 \times 540) + 0.2 \times 1 \times (100 - 30)$$

$$= (M + m) \times 1 \times (30 - 20)$$

$$\Rightarrow M + m = 12.2$$

Solving equations

$$m = 1.69 \text{ kg}$$

$$M = 10.48 \text{ kg}$$

6. **Ans (10.40 to 10.60)**

Using conservation of energy

$$(m \times 540) + m \times 1 \times (100 - 20)$$

$$= (m \times 80) + M \times 1 \times (20 - 0)$$

$$\Rightarrow m = 6.2 \text{ m}$$

$$(0.2 \times 540) + 0.2 \times 1 \times (100 - 30)$$

$$= (M + m) \times 1 \times (30 - 20)$$

$$\Rightarrow M + m = 12.2$$

Solving equations

$$m = 1.69 \text{ kg}$$

$$M = 10.48 \text{ kg}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (3)**

$$\Delta E_{2-1} = 13.6 \times z^2 \left[1 - \frac{1}{4} \right]$$

$$= 13.6 \times z^2 \left[\frac{3}{4} \right]$$

$$\Delta E_{3-2} = 13.6 \times z^2 \left[\frac{1}{4} - \frac{1}{9} \right]$$

$$= 13.6 \times z^2 \left[\frac{5}{36} \right]$$

$$\Delta E_{2-1} = \Delta E_{3-2} + 74.8$$

$$13.6 \times z^2 \left[\frac{3}{4} \right] = 13.6 \times z^2 \left[\frac{5}{36} \right] + 74.8$$

$$13.6 \times z^2 \left[\frac{3}{4} - \frac{5}{36} \right] = 74.8$$

$$z^2 = 9$$

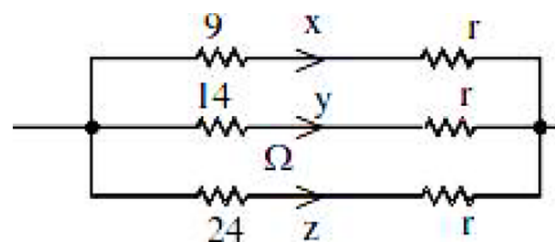
$$z = +3$$

8. **Ans (3)**

$$x(9+r) = y(14+r) \quad \text{_____ (1)}$$

$$xr = 7.5 \quad \text{_____ (2)}$$

$$yr = 5 \quad \text{_____ (3)}$$



By (1), (2), (3)

$$\Rightarrow x = 7.5 \text{ A} \Rightarrow y = 5 \text{ A}$$

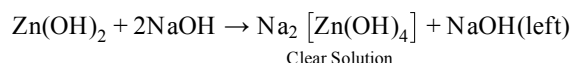
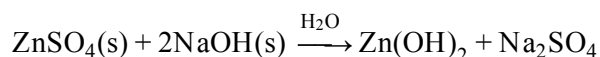
from (1) and (2) $r = 1 \Omega$

$$Z(24 + r) = x(9 + r)$$

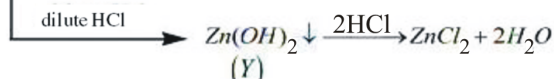
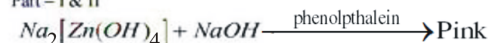
$$\Rightarrow z = 3 \text{ A}$$

$$V_3 = Zr = 3 \text{ V}$$

8. Ans (D)



Part - I & II



9. Ans (A)

Conceptual

10. Ans (A)

Nucleophile attacks less hindered carbon, steric

factors dominates electronic factors.

PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (330.00 to 342.00)

Volume of bulb (V) can be calculated as per

data given at 300K $PV = nRT$

$$\frac{114}{760} \times V = \frac{1}{100} \times 0.0821 \times 300$$

$$V = 1.642 \text{ lit.,}$$

Now at 600 K when NH_4Cl would have

decomposed $PV = nRT$

$$\frac{908}{760} \times 1.64 = n \times 0.082 \times 600$$

$$n = 0.039 \text{ moles}$$

$$\text{NH}_3 + \text{HCl} = 0.039 - 0.01 = 0.029 \text{ moles}$$

$$\text{NH}_3 = \frac{0.029}{2} = 0.0145 \text{ moles}$$

$$P_{\text{NH}_2} = \frac{0.0145}{0.039} \times 908 \approx 340 \text{ mm}$$

2. Ans (0.75 to 0.80)

Volume of bulb (V) can be calculated as per data

given at 300K $PV = nRT$

$$\frac{114}{760} \times V = \frac{1}{100} \times 0.0821 \times 300$$

$$V = 1.642 \text{ lit.,}$$

Now at 600 K when NH_4Cl would have

decomposed $PV = nRT$

$$\frac{908}{760} \times 1.64 = n \times 0.082 \times 600$$

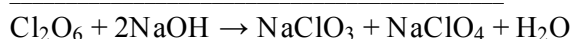
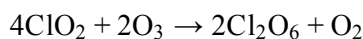
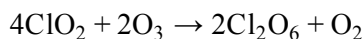
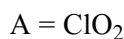
$$n = 0.039 \text{ moles}$$

$$\text{NH}_3 + \text{HCl} = 0.039 - 0.01 = 0.029 \text{ moles}$$

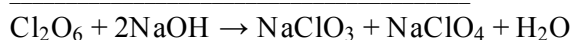
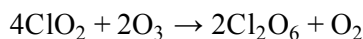
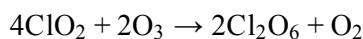
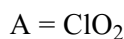
$$\text{NH}_3 = \frac{0.029}{2} = 0.0145 \text{ moles}$$

$$P_{\text{NH}_2} = \frac{0.0145}{0.039} \times 908 \approx 340 \text{ mm}$$

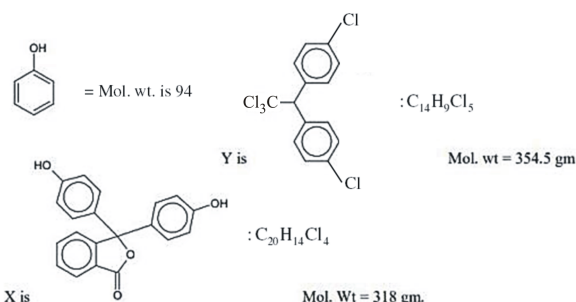
3. Ans (4.00)



4. Ans (2.00)



5. Ans (190.60 to 190.90)



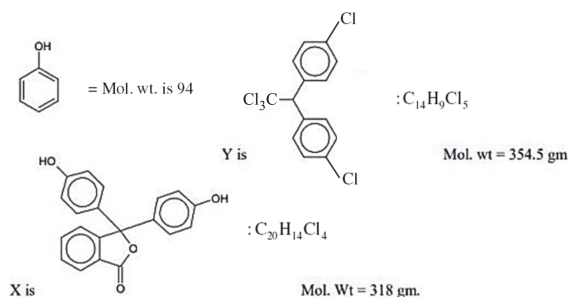
$$\text{Yield of X is : } \frac{1.5}{2} \times 318 \times \frac{80}{100} \text{ gm}$$

$$= \frac{954}{5} = 190.8 \text{ gms}$$

$$\text{Yield of Y is : } \frac{1.5}{2} \times 354.5 \times \frac{60}{100} \times \frac{90}{100}$$

$$= \frac{81 \times 3545}{2000} = \frac{287145}{2000} = 143.57 \text{ gms}$$

6. Ans (143.50 to 143.60)



$$\text{Yield of X is : } \frac{1.5}{2} \times 318 \times \frac{80}{100} \text{ gm}$$

$$= \frac{954}{5} = 190.8 \text{ gms}$$

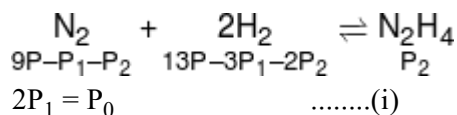
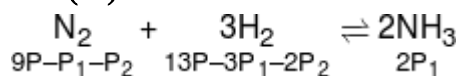
$$\text{Yield of Y is : } \frac{1.5}{2} \times 354.5 \times \frac{60}{100} \times \frac{90}{100}$$

$$= \frac{81 \times 3545}{2000} = \frac{287145}{2000} = 143.57 \text{ gms}$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (5)



$$13P - 3P_1 - 2P_2 = 2P_0 \text{(ii)}$$

$$P_{\text{eq}} = P_{\text{N}_2} + P_{\text{H}_2} + P_{\text{NH}_3} + P_{\text{N}_2\text{H}_4}$$

$$7P_0 = 22P - 2P_1 - 2P_2 \text{(iii)}$$

$$\text{On solving : } P = \frac{P_0}{2} \quad P_2 = \frac{3P_0}{2}$$

$$P_{\text{N}_2} = 9P - P_1 - P_2 = \frac{5P_0}{2}$$

$$P_{\text{H}_2} = 13P - 3P_1 - 2P_2 = 2P_0$$

$$P_{\text{NH}_3} = P_0$$

$$P_{\text{N}_2\text{H}_4} = \frac{3P_0}{2}$$

$$K_{P_1} = \frac{1}{20P_0^2} \quad K_{P_2} = \frac{3}{20P_0^2}$$

$$K_{P_1} + K_{P_2} = \frac{1}{20P_0^2} + \frac{3}{20P_0^2} = \frac{1}{5P_0^2}$$

$$= 5$$

8. Ans (1)

$$\Delta T_f = i \cdot k_f \cdot m$$

$$i = 3 [\because \text{co. No. } -6]$$

$$\text{So } [\text{CO}(\text{NH}_3)_5\text{Cl}] \text{Cl}_2$$

9. Ans (6)

Conceptual

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,D)

$$\int \underbrace{x}_{\text{I}} \cdot \underbrace{e^{x^2+x} \cdot (2x+1) dx}_{\text{II}} + \int e^{x^2+x} dx$$

$$x \cdot e^{x^2+x} - \int e^{x^2+x} dx + \int e^{x^2+x} dx$$

$$\therefore f(x) = xe^x$$

$$f'(x) = e^x(1+x) = 0 \Rightarrow x = -1$$

$$f''(x) = e^x(2+x)$$

$$f''(-1) > 0 \Rightarrow \text{Minima}$$

$$\therefore f(-1) = \frac{-1}{e} = m$$

$$\left[\frac{-1}{m} \right] = [e] = 2$$

3. Ans (A,B,D)

$$f(f(x)) = 1 - x$$

$$\Rightarrow f(f(f(x))) = 1 - f(x) \Rightarrow f(1 - x) + f(x) = 1$$

$$\therefore f\left(\frac{1}{4}\right) + f\left(\frac{3}{4}\right) = 1$$

$$J = \int_0^1 f(x) dx = \int_0^1 f(1-x) dx$$

$$\therefore 2J = \int_0^1 1 dx = 1 \Rightarrow J = \frac{1}{2}$$

$$f\left(\frac{1}{3}\right) + f\left(\frac{2}{3}\right) = 1 \Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\cos x}{(\sin x + \cos x)^3} dx$$

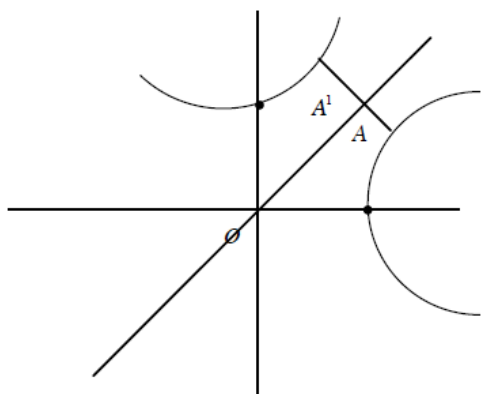
$$\therefore 2I = \int_0^{\frac{\pi}{2}} \frac{1}{(\sin x + \cos x)^2} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x} dx \quad \tan x = t$$

$$= \int_0^{\infty} \frac{\frac{dt}{1+t^2}}{1 + \frac{2t}{1+t^2}} = \int_0^{\infty} \frac{dt}{(t+1)^2}$$

$$= \left(-\frac{1}{t+1} \right)_0^{\infty} = (0+1) = 1 \Rightarrow \therefore I = \frac{1}{2}$$

5. Ans (B,C)



Clearly the curve are symmetric about $y = x$

$$A = 2A^1$$

$$2y \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2y} = 1$$

$$\Rightarrow y = \frac{1}{2}$$

$$\Rightarrow x = \frac{5}{4}$$

$$\therefore \left(\frac{5}{4}, \frac{1}{2}\right), \left(\frac{1}{2}, \frac{5}{4}\right) \text{ are points of contact}$$

$$\therefore \text{S.D.} = \sqrt{\frac{9}{16} + \frac{9}{16}}$$

$$= \sqrt{2 \cdot \frac{9}{16}}$$

$$= \frac{3}{4} \sqrt{2} = \frac{3}{2\sqrt{2}}$$

6. Ans (A,B,C,D)

$$f(2x) = 8x^3 - 6x + 1 = 2(4x^3 - 3x) + 1$$

$$f(2\cos\theta) = 2\cos 3\theta + 1$$

$$f\left(2\cos\frac{\pi}{9}\right) = 2\cos\frac{\pi}{3} + 1 = 0$$

$$f\left(2\cos\frac{4\pi}{9}\right) = 2\cos\frac{4\pi}{3} + 1 = 0$$

$$f\left(2\cos\frac{8\pi}{9}\right) = 2\cos\frac{8\pi}{3} + 1 = 0$$

$$f\left(2\sin\frac{\pi}{18}\right) = f\left(2\cos\frac{4\pi}{9}\right) = 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

$$a_1 = a-d, a_2 = a, a_3 = d$$

$$b_1 = \frac{b}{R}, b_2 = b, b_3 = bR$$

$$(a-d) \frac{b}{R} = 20, ab = 19,$$

$$(a+d) bR = 14$$

$$\left. \begin{aligned} \Rightarrow (19-bd) &= 20R \\ (19+bd) &= \frac{14}{R} \end{aligned} \right\} \Rightarrow 19R = 10R^2 + 7$$

$$\Rightarrow R = \frac{1}{2}, \frac{7}{5}$$

$$\text{when } R = \frac{7}{5}, a_4 b_4 = \frac{49}{25}$$

$$R = \frac{1}{2}, a_4 b_4 = 9.25$$

8. Ans (C)

$$a_1 = a-d, a_2 = a, a_3 = d$$

$$b_1 = \frac{b}{R}, b_2 = b, b_3 = b\pi$$

$$(a-d) \frac{b}{R} = 20, ab = 19,$$

$$(a+d) bR = 14$$

$$(19-bd) = 20R$$

$$(19+bd) = \frac{14}{R}$$

$$\Rightarrow 19R = 10R^2 + 7$$

$$\Rightarrow R = \frac{1}{2}, \frac{7}{5}$$

$$\text{when } R = \frac{7}{5}, a_4 b_4 = \frac{49}{25}$$

$$R = \frac{1}{2}, a_4 b_4 = 9.25$$

9. Ans (A)

Equation of plane containing P, A, B is

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 2 & 4 & 6 \\ -3 & 2 & -2 \end{vmatrix} = 0$$

$$\Rightarrow x + y - z = 0$$

10. Ans (B)

Let $(t - 2, 2t+3, 3t + 1)$ be the foot of perpendicular from A

$$\therefore 1(t-3)+2(2t+1)+3(3t-2)=0$$

$$\Rightarrow t = \frac{1}{2}$$

$$\therefore \text{Image of A} = (-4, 6, 2)$$

$$\therefore PA + PB = \sqrt{98} = 7\sqrt{2}$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (55.00)

$$\text{Tr}(3A_1) = \sum_{i=1}^3 \lambda^i \frac{\lambda^i}{\lambda^2} = 1+2+3$$

$$\text{Tr}(3^2.A_2) = \sum_{i=1}^3 \lambda^2 \frac{\lambda^i}{\lambda^4} = \frac{1+2+3}{3}$$

$$\therefore \text{Tr}(3^n.A_n) = \frac{1+2+3}{3^{n-1}}$$

$$\therefore L = \lim_{n \rightarrow \infty} 6 \left(1 + \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^{n-1}} \right)$$

$$= 6 \cdot \frac{3}{2} = 9$$

$$M = \text{Tr}(2B_1) = 2 \sum_{i=1}^3 \frac{2i}{2^2} = 1+2+3$$

$$\text{Tr}(2^2.B_2) = 2^2 \sum_{i=1}^3 \frac{2i}{2^4} = \frac{1+2+3}{2}$$

$$\therefore M = \lim_{n \rightarrow \infty} 6 \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} \right)$$

$$= 6 \times 2 = 12$$

$$f(x) = 1 - x^3,$$

$$g_n(x) = \frac{5}{\frac{1}{1-x^3} + \frac{1}{1-(2x)^3} + \dots + \frac{1}{1-(5x)^3}}$$

$$R = \lim_{x \rightarrow 0} \frac{1 - \frac{5}{\frac{1}{1-x^3} + \frac{1}{1-(2x)^3} + \dots + \frac{1}{1-(5x)^3}}}{x^3}$$

$$\therefore R = \lim_{x \rightarrow 0} \frac{\frac{1}{1-x^3} + \frac{1}{1-(2x)^3} + \dots + \frac{1}{1-(5x)^3} - 5}{x^3}$$

$$\lim_{x \rightarrow 0} \frac{1}{\left(\frac{1}{1-x^3} + \frac{1}{1-(2x)^3} + \dots + \frac{1}{1-(5x)^3} \right)}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{-(-3x^2)}{(1-x^3)^2} + \frac{(-24^8x^2)}{(1-(2x)^3)^2} - \frac{(81^{27}x^2)}{(1-(3x)^3)^2} + \frac{192^{64}x^2}{(1-(4x)^3)^2} + \frac{375^{125}x^2}{(1-(5x)^3)^2}}{3x^2} \cdot \frac{1}{5}$$

$$= 225 \times \frac{1}{5} = 45$$

$$\therefore 100 - R = 55$$

$$L + M + R = 9 + 12 + 45 = 66$$

2. Ans (66.00)

$$\text{Tr}(3A_1) = \sum_{i=1}^3 \lambda^i \frac{\lambda^i}{\lambda^2} = 1+2+3$$

$$\text{Tr}(3^2.A_2) = \sum_{i=1}^3 \lambda^2 \frac{\lambda^i}{\lambda^4} = \frac{1+2+3}{3}$$

$$\therefore \text{Tr}(3^n.A_n) = \frac{1+2+3}{3^{n-1}}$$

$$\therefore L = \lim_{n \rightarrow \infty} 6 \left(1 + \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^{n-1}} \right)$$

$$= 6 \cdot \frac{3}{2} = 9$$

$$M = \text{Tr}(2B_1) = 2 \sum_{i=1}^3 \frac{2i}{2^2} = 1+2+3$$

$$\text{Tr}(2^2.B_2) = 2^2 \sum_{i=1}^3 \frac{2i}{2^4} = \frac{1+2+3}{2}$$

$$\therefore M = \lim_{n \rightarrow \infty} 6 \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} \right)$$

$$= 6 \times 2 = 12$$

$$f(x) = 1 - x^3,$$

$$g_n(x) = \frac{5}{\frac{1}{1-x^3} + \frac{1}{1-(2x)^3} + \dots + \frac{1}{1-(5x)^3}}$$

$$R = \lim_{x \rightarrow 0} \frac{1 - \frac{5}{\frac{1}{1-x^3} + \frac{1}{1-(2x)^3} + \dots + \frac{1}{1-(5x)^3}}}{x^3}$$

$$\therefore R = \lim_{x \rightarrow 0} \frac{\frac{1}{1-x^3} + \frac{1}{1-(2x)^3} + \dots + \frac{1}{1-(5x)^3} - 5}{x^3}$$

$$\lim_{x \rightarrow 0} \frac{1}{\left(\frac{1}{1-x^3} + \frac{1}{1-(2x)^3} + \dots + \frac{1}{1-(5x)^3} \right)}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{-(-3x^2)}{(1-x^3)^2} + \frac{(-24^8x^2)}{(1-(2x)^3)^2} - \frac{(81^{27}x^2)}{(1-(3x)^3)^2} + \frac{192^{64}x^2}{(1-(4x)^3)^2} + \frac{375^{125}x^2}{(1-(5x)^3)^2}}{3x^2} \cdot \frac{1}{5}$$

$$= 225 \times \frac{1}{5} = 45$$

$$\therefore 100 - R = 55$$

$$L + M + R = 9 + 12 + 45 = 66$$

3. Ans (0.12 to 0.13)

$$A = 2, B = 4$$

$$\therefore f(x) = 12x^2 \int_0^1 yf(y) dy + 20 \int_0^1 xy^2f(y) dy + 4x$$

$$\text{Let } \alpha = \int_0^1 yf(y) dy, \beta = \int_0^1 y^2f(y) dy$$

$$\therefore f(x) = 12\alpha x^2 + 20\beta x + 4x$$

$$\therefore A = \int_0^1 (12\alpha x^3 + (20\beta + 4)x^2) dx$$

$$= \frac{12\alpha}{4} + \frac{(20\beta + 4)}{3}$$

$$2A + \frac{20B + 4}{3} = 0 \quad (1)$$

$$B = \int_0^1 (12Ay^4 + (20B + 4)x^3) dx$$

$$= \frac{12A}{5} + \left(\frac{20B + 4}{4} \right)$$

$$\frac{12A}{5} + 4B + 1 = 0$$

$$\therefore 3\alpha + 10\beta + 2 = 0$$

$$12\alpha + 20\beta + 5 = 0$$

$$\alpha = -\frac{1}{6}, 10\beta = -\frac{3}{2}$$

$$\Rightarrow \beta = -\frac{3}{20}$$

$$\therefore f(x) = -2x^2 + x$$

$$\therefore \text{Max value} = \frac{-1}{-8} = \frac{1}{8} = 0.125 \quad (0.12 \text{ to}$$

$$0.13)$$

4. Ans (0.04)

$$y = 0, y = -2x^2 + x$$

$$\therefore \text{Area} = \frac{1}{24} = 0.041$$

5. Ans (0.00)

$$P^n = 1 \text{ and } s = 1 + 5p + 9p^2 + \dots + (4n-3)p^{n-1}$$

$$ps = 1 + 5p^2 + \dots + (4n-3)p^n$$

$$(4n-3)p^n$$

$$(1-p)s = 1 + 4p + 4p^2 + \dots + 4p^{n-1} - (4n-3)p^n$$

$$= 1 + 4p(1 + p + \dots + p^{n-2}) - (4n-3)$$

$$= 1 + 4p \left(\frac{1 - p^{n-1}}{1 - p} \right) - (4n-3) = -4n$$

$$\Rightarrow s = \frac{4n}{p-1}$$

$$\therefore a = 4$$

$$p = \text{cis} \frac{2\pi}{n}$$

$$\Rightarrow 1 + 5 \cos \left(\frac{2\pi}{n} \right) + 9 \cos \left(\frac{4\pi}{n} \right) + \dots + (4n-3) \cos \left(\frac{2(n-1)\pi}{n} \right)$$

$$= \frac{4n}{\text{cis} \frac{2\pi}{n} - 1} = -2n - 2ni \cot \frac{\pi}{4}$$

Equating real and imaginary parts

$$b = -2, c = -2$$

$$\therefore a + b + c = 0$$

$$z = 4 - i(-4) = 4 + 4i$$

$$\therefore \arg. z = \frac{\pi}{4} = \theta$$

$$\Rightarrow 8\theta = 8 \times \frac{\pi}{4} = 2\pi$$

$$\therefore A = [6.28 \text{ to } 6.30]$$

6. Ans (6.28 to 6.30)

$$P^n = 1 \text{ and } s = 1 + 5p + 9p^2 + \dots + (4n-3)p^{n-1}$$

$$ps = 1 + 5p^2 + \dots + (4n-3)p^n$$

$$(1-p)s = 1 + 4p + 4p^2 + \dots + 4p^{n-1} - (4n-3)p^n$$

$$= 1 + 4p(1 + p + \dots + p^{n-2}) - (4n-3)p^n$$

$$= 1 + 4p \left(\frac{1-p^{n-1}}{1-p} \right) - (4n-3)p^n = -4n$$

$$\Rightarrow s = \frac{4n}{p-1}$$

$$\therefore a = 4$$

$$p = \text{cis} \frac{2\pi}{n}$$

$$\Rightarrow 1 + 5 \cos\left(\frac{2\pi}{n}\right) + 9 \cos\left(\frac{4\pi}{n}\right) + \dots + (4n-3) \cos\left(\frac{2(n-1)\pi}{n}\right)$$

$$= \frac{4n}{\text{cis} \frac{2\pi}{n} - 1} = -2n - 2ni \cot \frac{\pi}{4}$$

Equating real and imaginary parts

$$b = -2, c = -2$$

$$\therefore a + b + c = 0$$

$$z = 4 - i(-4) = 4 + 4i$$

$$\therefore \arg. z = \frac{\pi}{4} = \theta$$

$$\Rightarrow 8\theta = 8 \times \frac{\pi}{4} = 2\pi$$

$$\therefore A = [6.28 \text{ to } 6.30]$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (5)

$$g(\theta) = (t^2 + t) \frac{|\hat{a} \times \hat{b}|}{-(\hat{a} \cdot \hat{b})^2}$$

$$= |\hat{a} \times \hat{b}|^4 + |\hat{a} \times \hat{b}|^2 - (\hat{a} \cdot \hat{b})^4 + (\hat{a} \cdot \hat{b})^2$$

$$= \sin^4 \theta + \sin^2 \theta - \cos^4 \theta + \cos^2 \theta$$

$$= 1 - \cos 2\theta = 2\sin^2 \theta$$

$$\therefore 2g(\theta) = 1 \Rightarrow 4\sin^2 \theta = 1$$

$$\Rightarrow \sin^2 \theta = \frac{1}{4}$$

$$\therefore \left[\hat{a} \cdot \hat{a} \times \hat{b} \cdot \hat{a} \times (\hat{a} \times \hat{b}) \right]$$

$$= |\hat{a} \times (\hat{a} \times \hat{b})|^2 = 1 (\hat{a} \times \hat{b}) = \sin^2 \theta$$

$$\therefore \frac{p}{q} = \frac{1}{4}$$

$$\therefore p + q = 5 \text{ is least}$$

8. Ans (4)

$$N = 1.3.5.7.9.11.13. \frac{1}{1} \cdot \frac{1}{3} \cdot \frac{1}{5} \cdot \dots \cdot \frac{1}{13} \log_2 3. \log_3 4. \dots \log_{15} 16 = 4$$

$$\tan^2 x + \cot^2 x + 2 = 4 \sin^2 2x$$

$$\text{L.H.S} \geq 4, \quad \text{R.H.S.} \geq 4$$

$$\therefore \tan^2 x = \cot x = 1 = \sin^2 2x$$

$$\therefore \text{No. of solutions} = 4$$

9. Ans (8)

$$f(f(x)) = (x + x^2 + x^4 + x^8)$$

$$+ (x + x^2 + x^4 + x^8)^2 + (x + x^2 + x^4 + x^8)^4$$

$$+ (x + x^2 + x^4 + x^8)^8 + (x + x^2 + x^4 + x^8)^{16} + \dots$$

$$\therefore \text{Required coefficient} = 2 + 4C_1 + 4C_2 + 8C_2 = 40$$